

Project Title

High-Performance Text Searching using KMP Algorithm

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Describe what your project presents within 150 words.

This project presents a High-Performance Text Searching Application using the Knuth–Morris–Pratt (KMP) algorithm for efficient exact pattern matching in large text files.

Traditional brute-force searching repeatedly compares characters, which can result in unnecessary comparisons and reduced performance for large datasets. The proposed application overcomes this limitation by preprocessing the search pattern and constructing a prefix table, also known as the failure function.

This enables the KMP algorithm to skip unnecessary comparisons and locate all occurrences of a given pattern efficiently. The system supports searching across multiple text files and identifies the matching positions within the files.

It also provides visualization of the detected matches, making the searching process easier to understand and analyze. The application demonstrates the linear-time efficiency of KMP and its practical usefulness in applications such as log analysis, plagiarism detection, text mining, and document analysis.

The application provides a user-friendly interface that allows users to enter patterns and analyze search results easily. It reduces redundant character comparisons and improves the overall efficiency of text-searching operations.

The project also demonstrates scalability for large text datasets, showing how efficient pattern-matching techniques can be applied to real-world text-processing problems.

Signature of the Guide